

Problem 3

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Problem 1 Find an equation of the tangent line at the given point. Then graph the function and the tangent line on the same plot. $f(x) = \sqrt{x+1}$ at $x = 3$

Explanation. The slope of tangent line is given by: $f'(3) = \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$.
This limit has form $\frac{0}{0}$, so it is an indeterminate form.

$$\begin{aligned}
 f'(3) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{3+h+1} - \sqrt{4}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - \sqrt{4}}{h} \cdot \frac{\sqrt{4+h} + \sqrt{4}}{\sqrt{4+h} + \sqrt{4}} \\
 &= \lim_{h \rightarrow 0} \frac{4+h-4}{h(\sqrt{4+h} + \sqrt{4})} \\
 &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{4+h} + 2)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{4+h} + 2} \\
 &= \frac{1}{\sqrt{4+0} + 2} \\
 &= \frac{1}{4}
 \end{aligned}$$

Point on tangent line: $(3, f(3)) = (3, 2)$.

Equation of tangent line:

$$\begin{aligned}
 y - f(3) &= \frac{1}{4}(x - 3) \implies y - 2 = \frac{1}{4}x - \frac{3}{4} \\
 &\implies y = \frac{1}{4}x + 1\frac{1}{4}.
 \end{aligned}$$

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